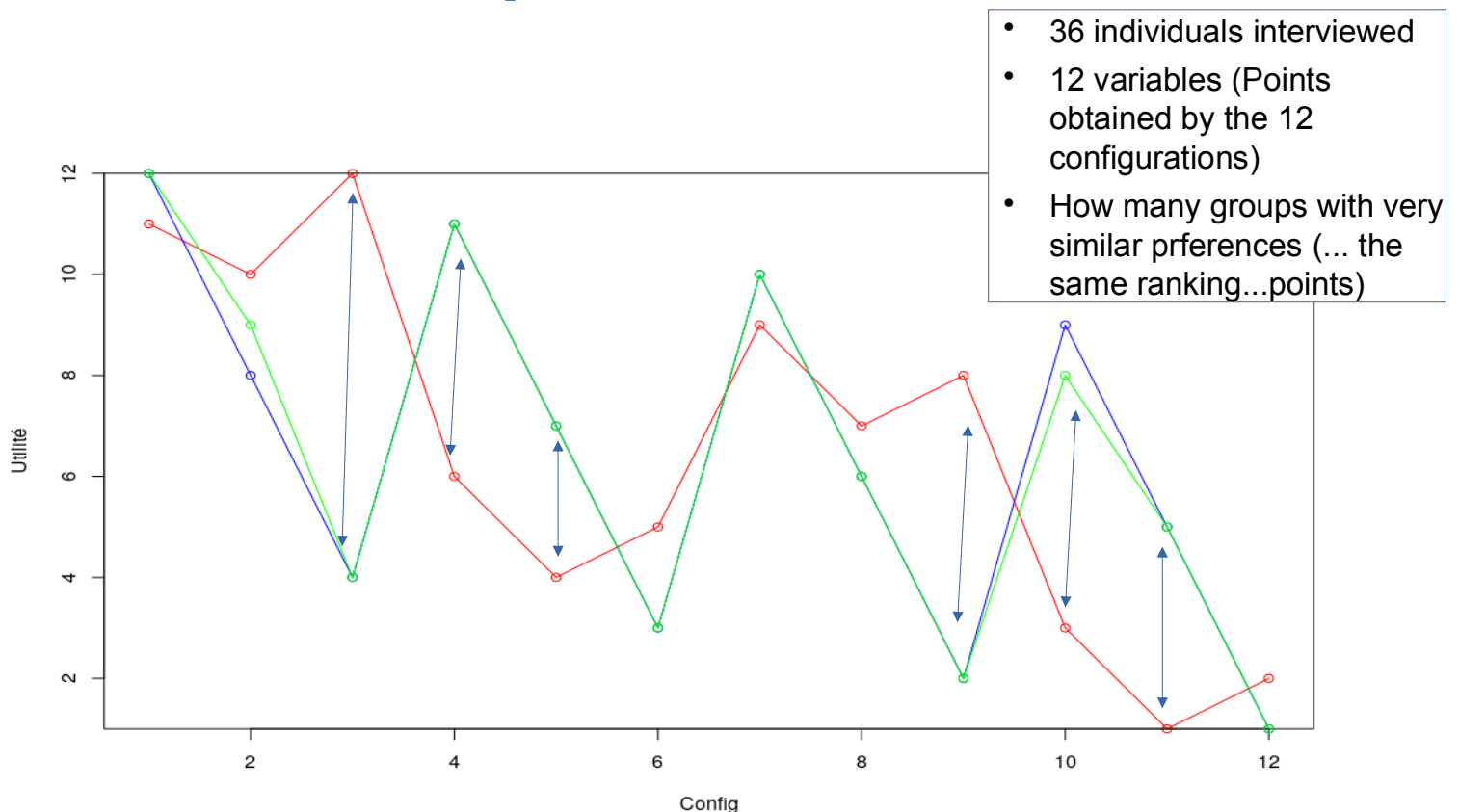


Hierarchical Agglomerative Clustering (HAC)

Segmentation Needs and Preferences

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Do the customers have the same preferences?



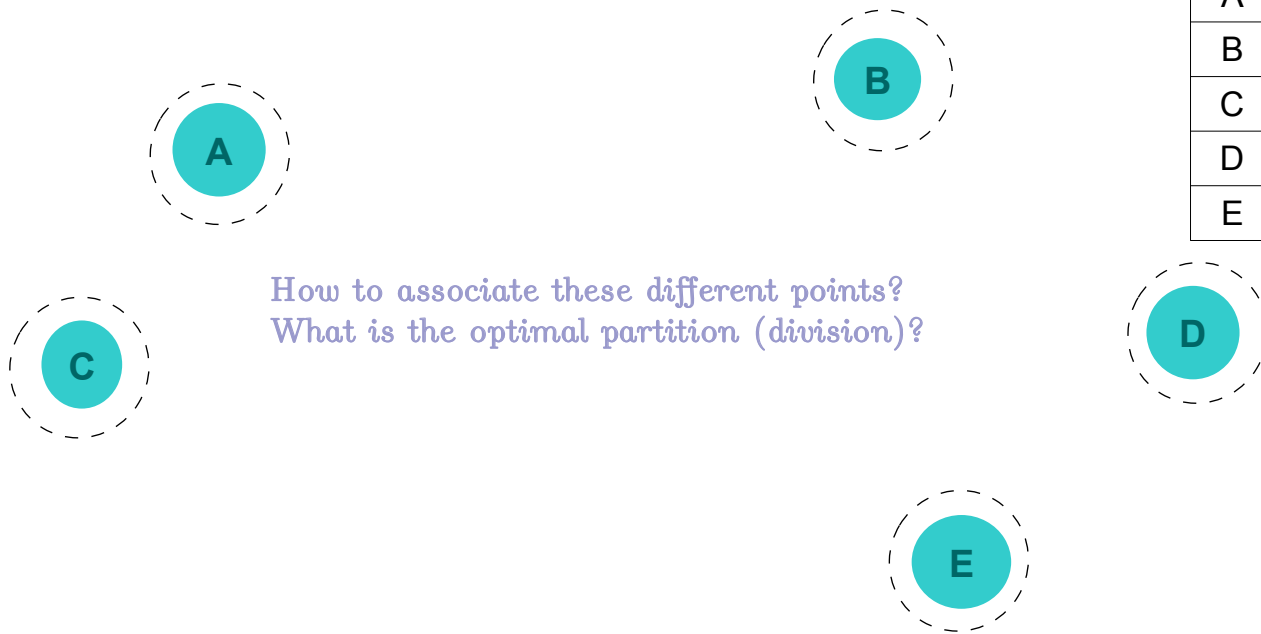
48

Classification (...grouping) problem

$n=5$ (5 points)
 $p=2$ (2 dimensions)

Individual(x,y)

	X	Y
A	0.5	2
B	4.5	2.5
C	0	1
D	6	1
E	5	0



49

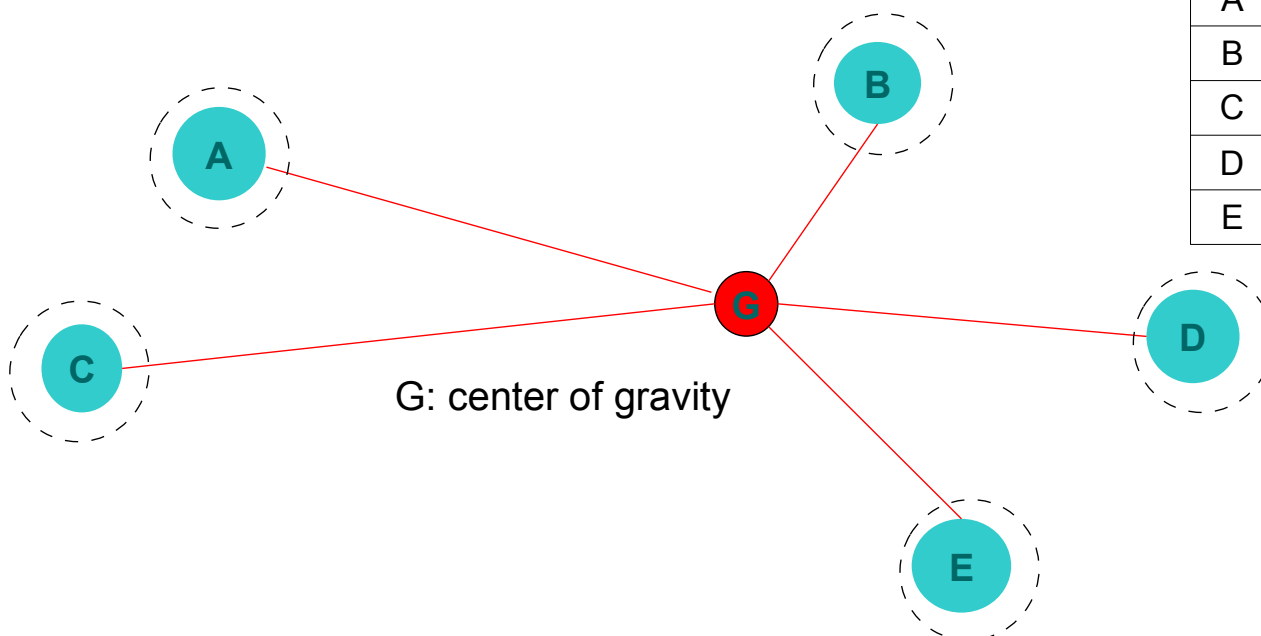
Ascending Hierarchical Classification

Inter-class inertia: between-clusters sum of squares (BCSS)

Intra-class inertia: within-clusters sum of squares (WCSS)

Individual(x,y)

	X	Y
A	0.5	2
B	4.5	2.5
C	0	1
D	6	1
E	5	0



50

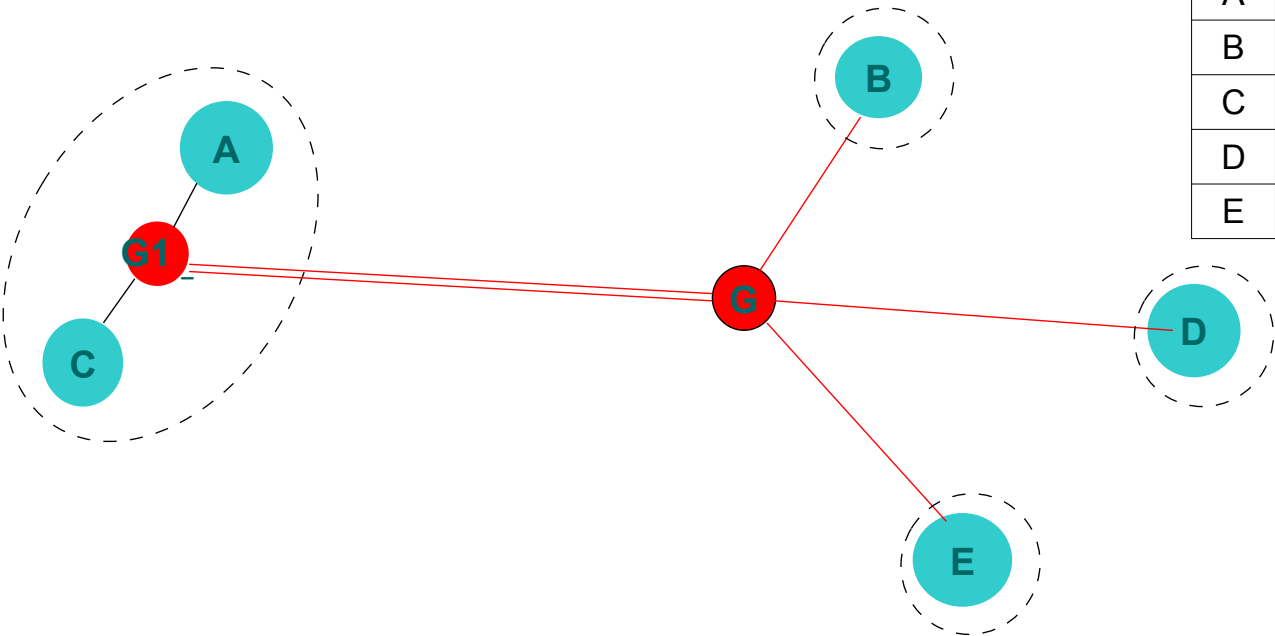
Ascending Hierarchical Classification

Graphic Illustration

Inter-class inertia: between-clusters sum of squares (BCSS)
Intra-class inertia: within-clusters sum of squares (WCSS)

Individual(x,y)

	X	Y
A	0.5	2
B	4.5	2.5
C	0	1
D	6	1
E	5	0



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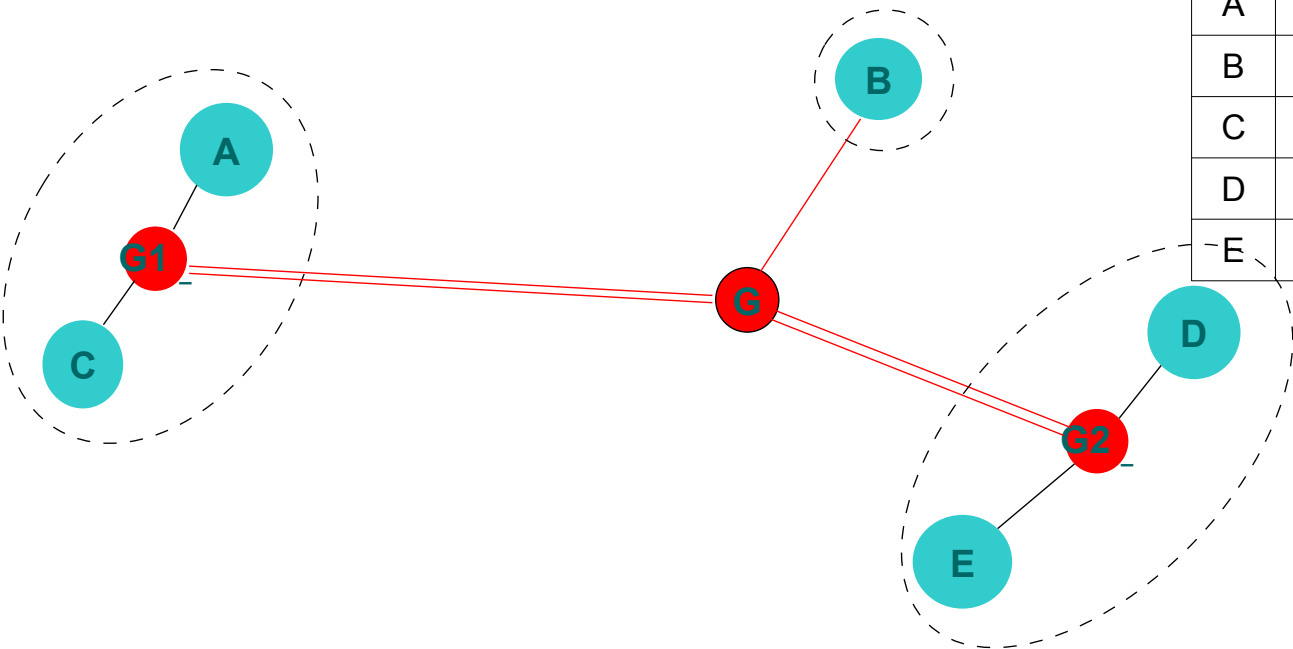
Ascending Hierarchical Classification

Graphic Illustration

Inter-class inertia: between-clusters sum of squares (BCSS)
Intra-class inertia: within-clusters sum of squares (WCSS)

Individual(x,y)

	X	Y
A	0.5	2
B	4.5	2.5
C	0	1
D	6	1
E	5	0



52

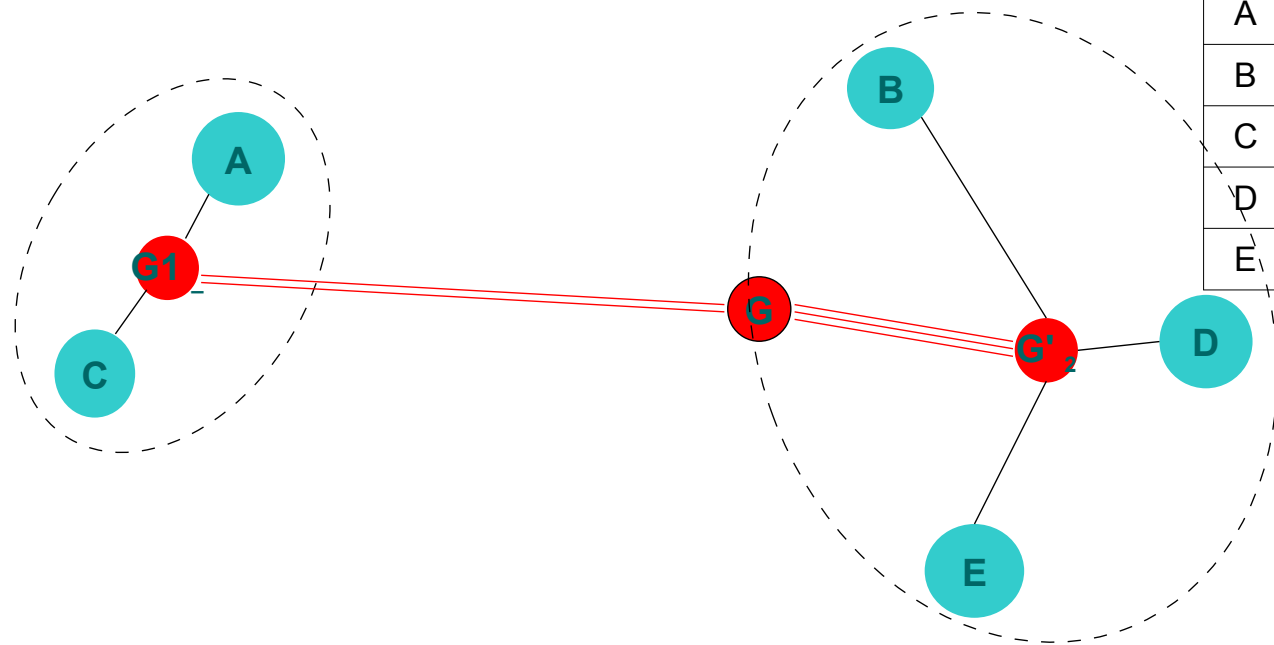
Ascending Hierarchical Classification

Graphic Illustration

Inter-class inertia: between-clusters sum of squares (BCSS)
Intra-class inertia: within-clusters sum of squares (WCSS)

Individual(x,y)

	X	Y
A	0.5	2
B	4.5	2.5
C	0	1
D	6	1
E	5	0



53

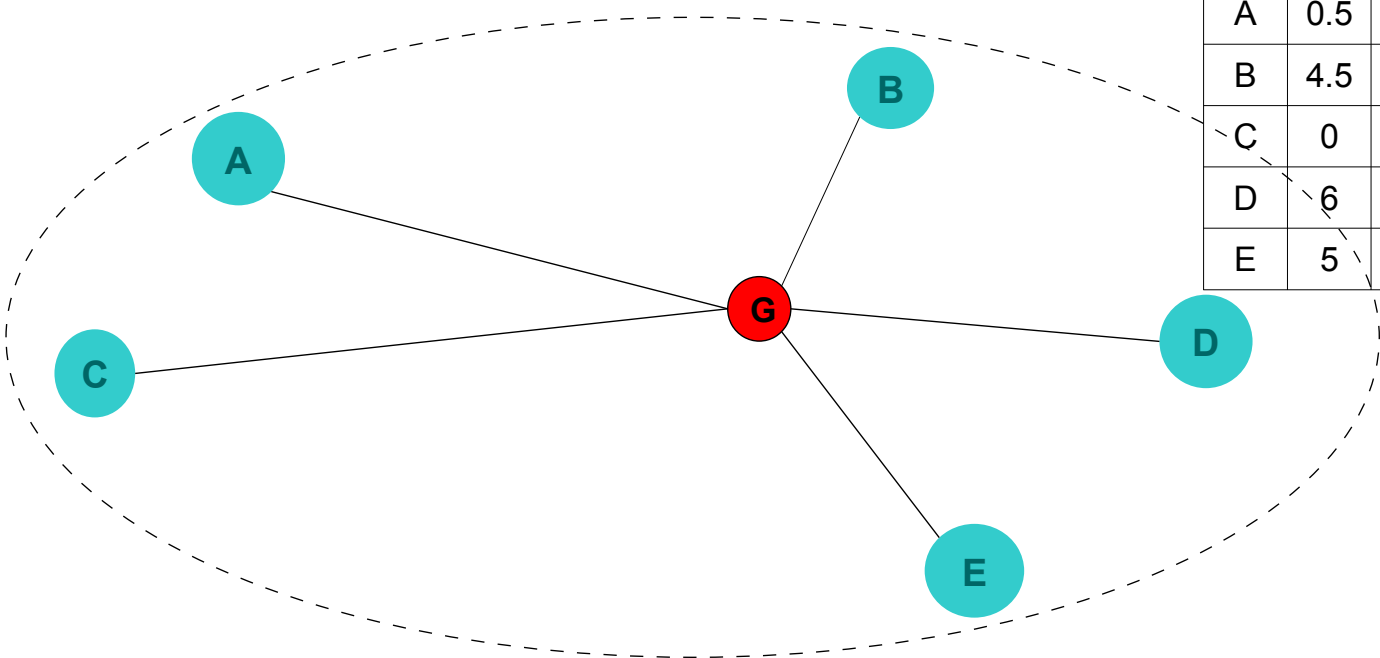
Ascending Hierarchical Classification

Graphic Illustration

Inter-class inertia: between-clusters sum of squares (BCSS)
Intra-class inertia: within-clusters sum of squares (WCSS)

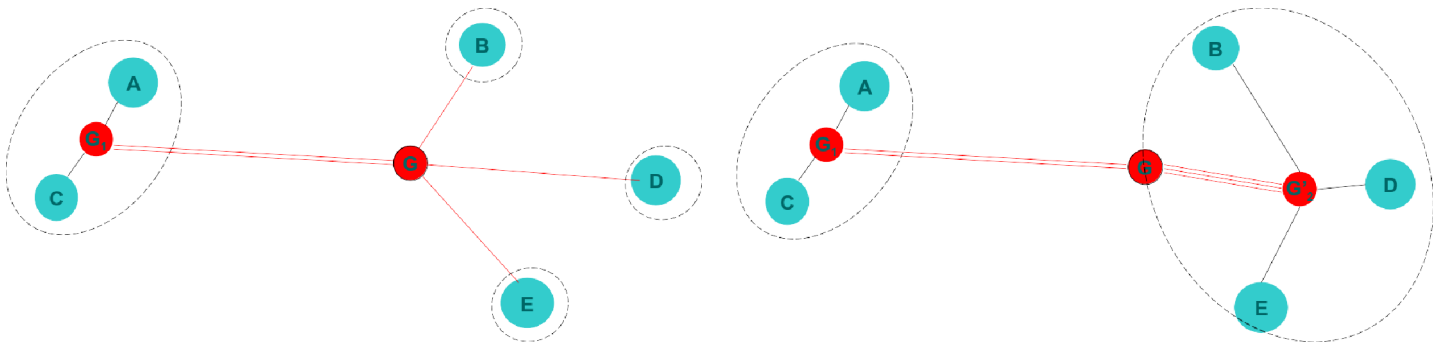
Individual(x,y)

	X	Y
A	0.5	2
B	4.5	2.5
C	0	1
D	6	1
E	5	0



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Inertia Variations



- At the start: **Total Inertia** = **Inter-class Inertia** (between-clusters variance)
- At the end: **Total Inertia** = **Intra-class Inertia** (within-clusters variance)
- At each step, **Total Inertia** = **Inter-class Inertia** + **Intra-class Inertia**

Follow the increase of intra-class inertia

< === >

Track the decrease in inter-class inertia

- Differences between classes gradually turn into differences between objects within classes

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Total Inertia

- Let X be a cloud of points corresponding to the description of n objects using p variables

$$X = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix}$$

- g is the center of gravity of the point cloud whose coordinates are the averages of the different variables:

$$G = \begin{pmatrix} \bar{x}_1 \\ \bar{x}_2 \\ \vdots \\ \bar{x}_p \end{pmatrix}$$

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Total inertia

- The dispersion around its center of gravity, of the cloud N formed by the set of n points, is measured using the total inertia of the cloud defined by:

$$G = \begin{pmatrix} \bar{X}_1 \\ \bar{X}_2 \\ \dots \\ \bar{X}_p \end{pmatrix} \qquad I(N, g) = \frac{1}{n} \sum_{i=1}^n d^2(X_i, g)$$

- where $d^2(X_i, g)$ is the square of the distance between the i-th point (X_i) and the center of gravity g.

$$d^2(X_i, g) = \sum_{j=1}^p (x_{ij} - \bar{X}_j)^2$$

- Inertia is akin to “ a measure of differences ”

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Total Inertia

- The total inertia can be calculated directly as the sum of the variances of the variables of the problem

$$\begin{aligned} I(N, g) &= \frac{1}{n} \sum_{i=1}^n d^2(X_i, g) \\ &= \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^p (x_{ij} - \bar{X}_j)^2 \\ &= \sum_{j=1}^p \frac{1}{n} \sum_{i=1}^n (x_{ij} - \bar{X}_j)^2 \\ &= \sum_{j=1}^p \sigma_j^2 \end{aligned}$$

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Ascending Hierarchical Classification Formalization

- Objective: Group objects into classes (groups)
 - Objects look alike within a class
 - Objects are very different if they are not associated within a class.
- The classification algorithm is iterative. At each stage, we group the 2 closest classes
 - Step 0: There are as many classes as individuals
 - Final Stage: There is only one class left
- At each step, the loss of inter-class inertia is calculated.
 - When this loss increases sharply, the grouping leads to associating 2 classes that are not very close.
 - The optimal partition corresponds to the groupings defined in the previous step.

$$\begin{aligned}\Delta Inertia_{Inter-clases} &= I(G_1, G_2, \dots, G_i, \dots, G_j, \dots, G_k) \\ &\quad - I(G_1, G_2, \dots, G_i \cup G_j, \dots, G_k) \\ &= \frac{n_i n_j}{n(n_i + n_j)} d^2(g_i, g_j)\end{aligned}$$

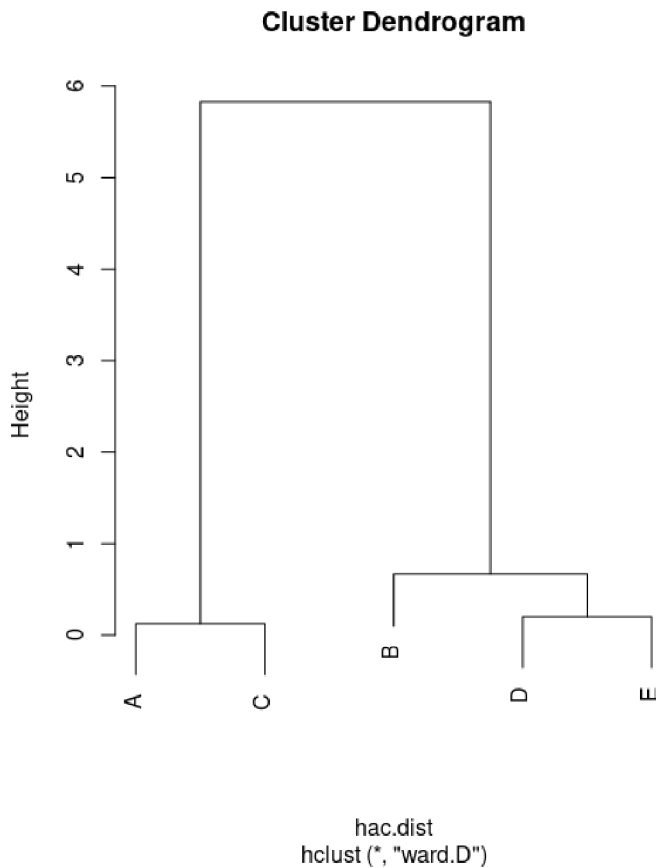
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Ascending Hierarchical Classification

```
# *****
# Agglomerative Hierarchical Clustering: Introductory Example
# *****
# Loading the necessary packages
library("readxl")
# Initialization (clearing the entire workspace)
rm(list=ls())
# Loading the data
data_hac <- read_excel("MDA_data.xlsx", sheet = "HAC")
# Converting data to a data frame and setting row names
data_hac <- data.frame(data_hac, row.names = 1)
# Calculating distances between different data points
hac.dist <- dist(data_hac, method = "euclidean")^2 / (2 * nrow(data_hac))
print(hac.dist)
# Implementing agglomerative hierarchical clustering
# based on the distance matrix
hac.res <- hclust(hac.dist, method = "ward.D")
# Displaying the within-cluster loss of inertia
# in numeric form
print(hac.res$height)
# Displaying the within-cluster loss of inertia
# as a barplot
barplot(hac.res$height)
# Displaying the dendrogram
plot(hac.res)
# History of cluster merging
hac.res$merge
# Cutting the dendrogram into 'k' clusters (in this case, k=3)
seg <- cutree(hac.res, k = 3)
# Adding cluster membership to the original data
data_hac <- cbind(data_hac, seg)
# Calculating the number of observations in each cluster
# Here, it represents the number of individuals in each segment
table(seg)
```

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The dendrogram

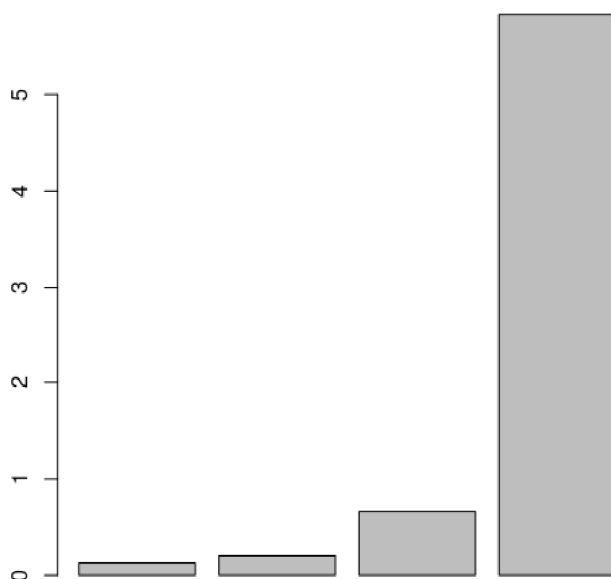


- The dendrogram presents the chronology of the associations made as well as their consequences in terms of “loss of inter-class inertia. »
- The “height” measurement (0.125;0.20;0.66;5.82)

```
# Affichage du dendrogramme  
plot(hac.res, labels=data_hac$Id)
```

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Evolution of intra-class inertia during mergers



This representation presents the “height” variable to each of the associations made

```
# Evolution des pertes d'inertie inter-classes  
# sous forme chiffrée  
print(hac.res$height)  
# sous forme d'histogramme  
barplot(hac.res$height)
```

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Calculation of inter-class inertia loss (Ward index)

- We have k groups $G_1, G_2, \dots, G_k$.
- In the next step, we decide to modify the typology by grouping the groups G_i and G_j . We replace G_i and G_j by the union $G_i \cup G_j$.
- The decrease in inter-class inertia (or the increase in intra-class inertia) is equal to:

$$\begin{aligned} \Delta Inertie_{Inter-classes} &= I(G_1, G_2, \dots, G_i, \dots, G_j, \dots, G_k) \\ &\quad - I(G_1, G_2, \dots, G_i \cup G_j, \dots, G_k) \\ &= \frac{n_i n_j}{n(n_i + n_j)} d^2(g_i, g_j) \end{aligned}$$

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Back to Conjoint Analysis

```
Call:
lm(formula = meansPts ~ data_ac$Ecran_17 + data_ac$HD_2To + data_ac$Prix_500 +
    data_ac$Prix_600)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-0.15741 -0.06655  0.01389  0.05787  0.21296
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.12037    0.08630   12.98 3.74e-06 ***
data_ac$Ecran_17 2.28704    0.07719   29.63 1.29e-08 ***
data_ac$HD_2To   2.52778    0.07719   32.75 6.41e-09 ***
data_ac$Prix_500 5.68750    0.09454   60.16 9.20e-11 ***
data_ac$Prix_600 3.22917    0.09454   34.16 4.78e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

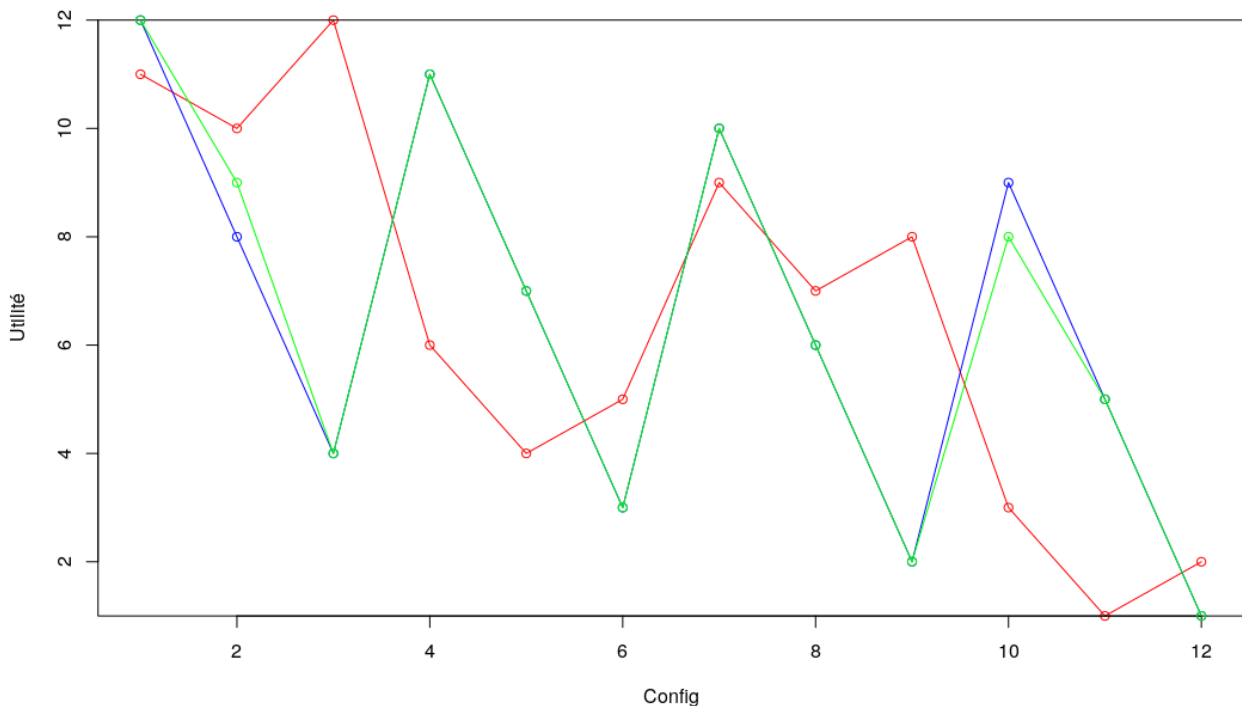
Residual standard error: 0.1337 on 7 degrees of freedom
Multiple R-squared:  0.9987,    Adjusted R-squared:  0.998
F-statistic: 1398 on 4 and 7 DF, p-value: 3.107e-10
```

For an equivalent cost, is it more interesting to:

- Propose a *larger screen or HD* ?
- Offer a big screen and a big HD or a *lower price of 200€* ?
-

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Back to Conjoint Analysis



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Application to the Laptop Case

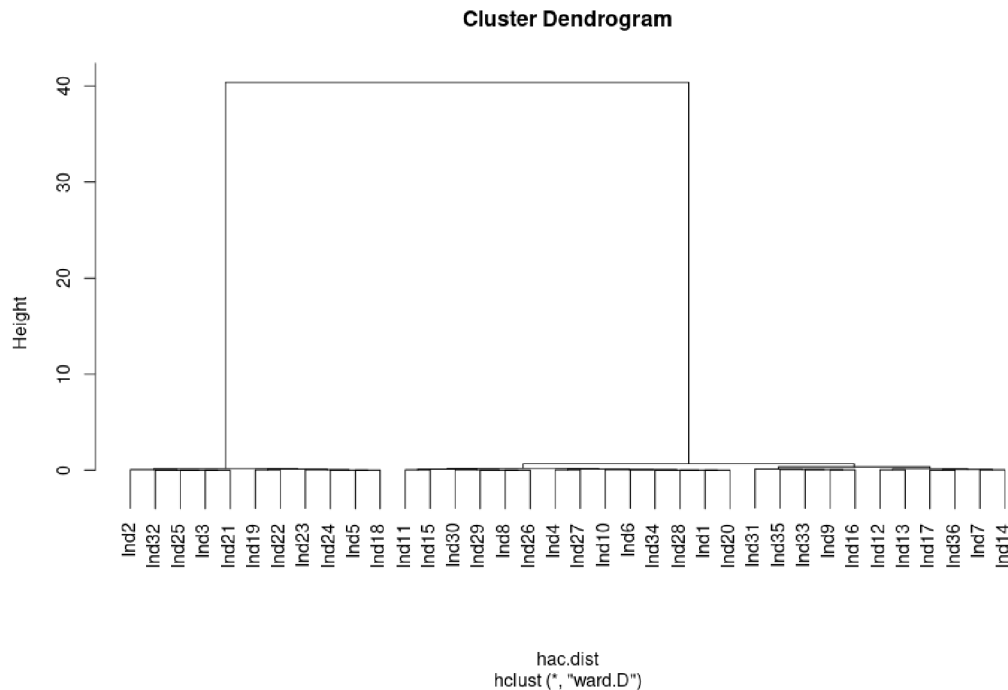
- In your opinion, how many consumer segments are present in the studied laptop market?
- How can you use the conjoint analysis in this case?
- what are the specific preferences of the segments studied?

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Application: Conjoint analysis

Continuation and end

```
> print(hac.res$height)
[1] 0.00000000 0.00000000 0.00000000 0.00000000 0.00000000 0.00000000 0.00000000 0.00000000
[9] 0.00000000 0.00000000 0.00000000 0.02777778 0.02777778 0.02777778 0.02777778 0.02777778
[17] 0.02777778 0.04166667 0.04166667 0.04444444 0.04444444 0.04629630 0.04814815 0.06481481
[25] 0.06944444 0.09722222 0.10555556 0.10648148 0.11574074 0.15277778 0.17356902 0.18849206
[33] 0.36161616 0.67155844 40.38223345
```

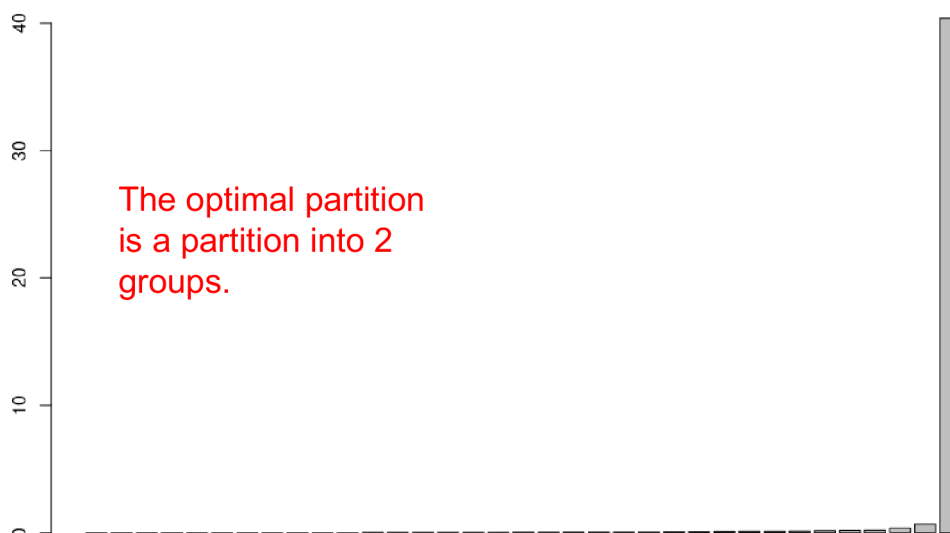


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Application: Conjoint analysis

Continuation and end

```
> print(hac.res$height)
[1] 0.00000000 0.00000000 0.00000000 0.00000000 0.00000000 0.00000000 0.00000000 0.00000000
[9] 0.00000000 0.00000000 0.00000000 0.02777778 0.02777778 0.02777778 0.02777778 0.02777778
[17] 0.02777778 0.04166667 0.04166667 0.04444444 0.04444444 0.04629630 0.04814815 0.06481481
[25] 0.06944444 0.09722222 0.10555556 0.10648148 0.11574074 0.15277778 0.17356902 0.18849206
[33] 0.36161616 0.67155844 40.38223345
```



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Application: Conjoint analysis

Continuation and end

```
# Découpage des groupes en fonction de l'évolution du parametre height et du dendrogramme
# 2 segments constitués
seg<-cutree(hac.res,k=2)
# Constitution d'un tableau ajout de la variable "seg" (segment d'appartenance de chaque individu)
Pts<-cbind(Pts,seg)
# Nombre d'individus par segment
table(seg)
```

```
> table(seg)
seg
 1  2
25 11
```

	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	seg
Ind1	1	5	9	2	6	10	4	7	11	3	8	12	1
Ind2	3	2	1	7	9	8	4	6	5	10	12	11	2
Ind3	2	3	1	7	9	8	4	6	5	10	12	11	2
Ind4	1	5	9	2	7	10	4	6	11	3	8	12	1
Ind5	1	3	2	7	9	8	4	6	5	10	12	11	2
Ind6	1	5	9	3	6	10	4	7	11	2	8	12	1
Ind7	2	4	9	1	6	10	3	7	11	5	8	12	1
Ind8	1	5	9	2	6	10	3	7	11	4	8	12	1
Ind9	2	5	9	3	6	10	1	7	11	4	8	12	1
Ind10	1	5	8	2	6	10	4	7	11	3	9	12	1
Ind11	1	5	9	2	6	10	3	8	11	4	7	12	1
Ind12	1	4	9	2	5	10	3	7	11	6	8	12	1

Application: Conjoint analysis

Continuation and end

```
# Création d'une tableau de résultats agrégeant les estimations des coefficients estimées
# Résultats obtenus avec tous les individus
coeff<-matrix(reg$coefficients, nrow=1)
```

```
# Estimation réalisée avec les seuls individus du groupe 1
meansALL<-colMeans(Pts[seg== 1,1:12])
print(meansALL)
# Estimation réalisée avec les seuls individus du groupe 1
reg<-lm(meansALL~data_ac$Ecran_17+data_ac$HD_2To+data_ac$Prix_500+data_ac$Prix_600)
summary(reg)
# Résultats obtenus avec les individus du groupe 1
coeff<-rbind(coeff,reg$coefficients)
```

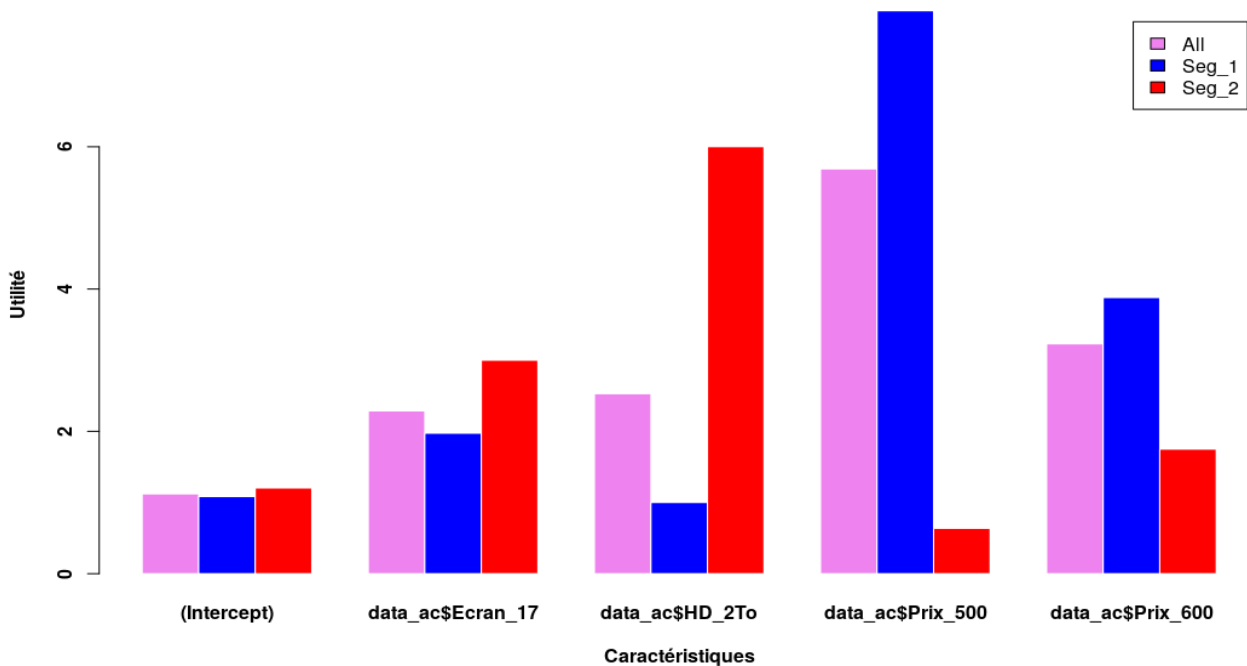
```
# Estimation réalisée avec les seuls individus du groupe 2
meansALL<-colMeans(Pts[seg== 2,1:12])
print(meansALL)
# Estimation réalisée avec les seuls individus du groupe 2
reg<-lm(meansALL~data_ac$Ecran_17+data_ac$HD_2To+data_ac$Prix_500+data_ac$Prix_600)
summary(reg)
# Résultats obtenus avec les individus du groupe 2
coeff<-rbind(coeff,reg$coefficients)
row.names(coeff)<-c("All","Seg_1","Seg_2")
# Présentation de l'ensemble des résultats (Tous les ind. puis par segment)
# sous forme chiffrée
print(coeff)
```

```
> print(coeff)
      (Intercept) data_ac$Ecran_17 data_ac$HD_2To data_ac$Prix_500 data_ac$Prix_600
[1,]  1.120370      2.287037      2.527778      5.6875000      3.229167
[2,]  1.083333      1.973333      1.000000      7.9100000      3.880000
[3,]  1.204545      3.000000      6.000000      0.6363636      1.750000
```

Application: Conjoint analysis

Continuation and end

```
# sous forme graphique
barplot(coeff, col=colors()[c(640,26,552)] , border="white", font.axis=2, beside=T,
        legend=rownames(coeff), xlab="Caractéristiques", ylab = "Utilité",font.lab=2)
```



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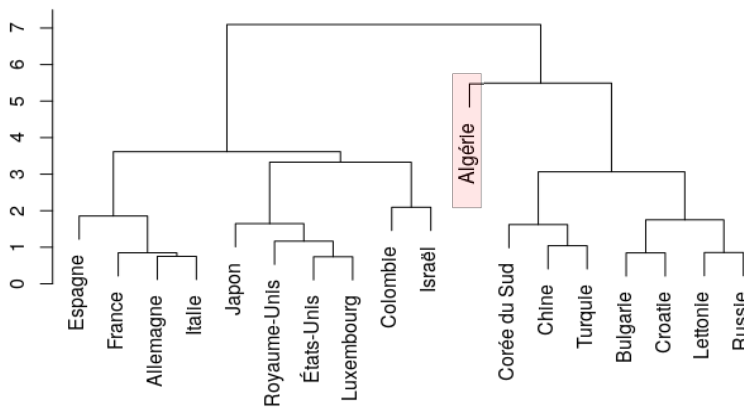
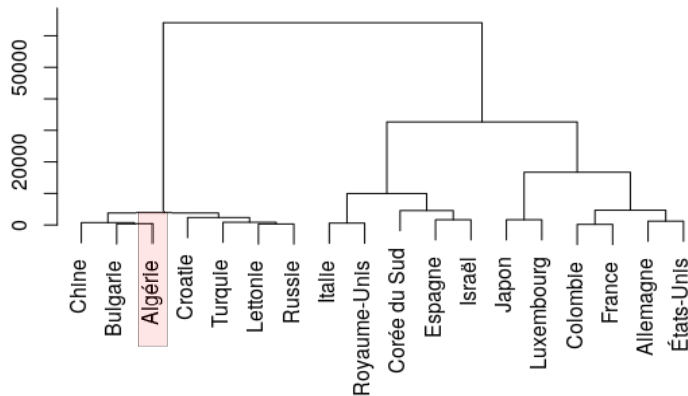
Back to the HAC Some difficulties...

- Dependencies on chosen measurement units.
the variables must be **normalized (center and reduce)!**
- Importance of the number of representative variables of different dimensions
 - ~ Demography, Economy, ...
 - => update of independent dimensions by applying Principal Component Analysis.

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HAC with and without normalization

▲	Pays	T%Ch	PNB/H	EspV	IndFec
1	Allemagne	12.7	28230	76.74	1.35
2	Bulgarie	13.7	1170	70.89	1.11
3	Chine	3.0	710	69.86	1.90
4	Colombie	12.1	25620	70.29	2.74
5	Corée du Sud	2.6	11390	72.59	1.60
6	Croatie	16.5	4640	72.76	1.45
7	Espagne	20.8	14530	77.91	1.16
8	États-Unis	5.0	29440	76.51	2.00
9	France	12.3	25430	78.31	1.75
10	Israël	7.7	16190	77.81	2.55
11	Italie	12.3	20180	78.26	1.23
12	Japon	3.4	38260	80.54	1.39
13	Lettonie	7.4	2300	69.66	1.09
14	Luxembourg	3.6	36600	76.66	1.71
15	Royaume-Unis	5.7	20760	77.17	1.71
16	Russie	10.8	2610	66.96	1.24
17	Turquie	6.9	3200	69.26	2.45
18	Algérie	28.7	1530	70.58	3.48



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Reduced centered variables

$$x_i^C = x_i - \bar{X} \quad x_i^{CR} = \frac{x_i - \bar{X}}{\sigma_x}$$

	Données Brutes		Données centrées		Données centrées et réduites	
	Poids	Taille	Poids_C	Taille_C	Poids_CR	Taille_CR
	60,00	155,00	-17,45	-17,90	-1,41	-1,94
	75,00	180,00	-2,45	7,10	-0,20	0,77
	61,00	162,00	-16,45	-10,90	-1,33	-1,18
	76,00	175,00	-1,45	2,10	-0,12	0,23
	64,00	157,00	-13,45	-15,90	-1,08	-1,72
	78,00	173,00	0,55	0,10	0,04	0,01
	67,00	170,00	-10,45	-2,90	-0,84	-0,31
	80,00	175,00	2,55	2,10	0,21	0,23
	68,00	164,00	-9,45	-8,90	-0,76	-0,96
	85,00	179,00	7,55	6,10	0,61	0,66
	69,00	162,00	-8,45	-10,90	-0,68	-1,18
	90,00	175,00	12,55	2,10	1,01	0,23
	70,00	169,00	-7,45	-3,90	-0,60	-0,42
	96,00	180,00	18,55	7,10	1,49	0,77
	70,00	170,00	-7,45	-2,90	-0,60	-0,31
	96,00	185,00	18,55	12,10	1,49	1,31
	72,00	178,00	-5,45	5,10	-0,44	0,55
	98,00	189,00	20,55	16,10	1,66	1,74
	73,00	173,00	-4,45	0,10	-0,36	0,01
	101,00	187,00	23,55	14,10	1,90	1,53
Moyenne :	77,45	172,90	0,00	0,00	0,00	0,00
Variance :	154,05	85,19	154,05	85,19	1,00	1,00
Ecart-type :	12,41	9,23	12,41	9,23	1,00	1,00

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To review and summarize

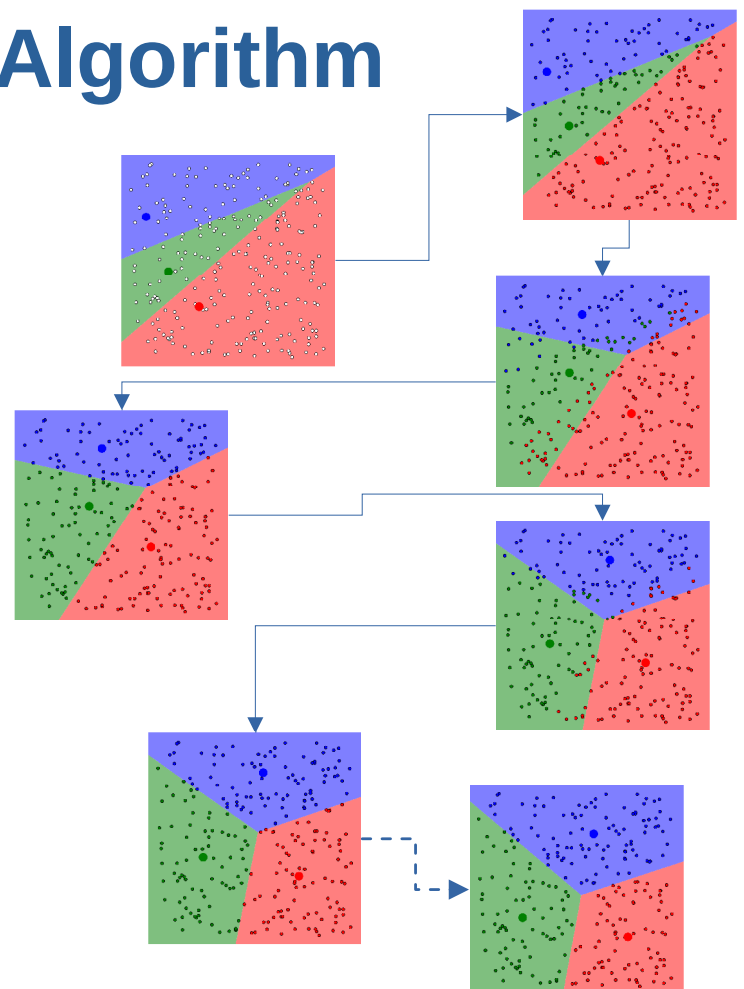
- What is the general objective of performing a classification ?
- How should objects be associated within a group ?
- How should groups relate to each other ?
- What are the indicators used to define these two concepts when applying agglomerative hierarchical clustering ?
- Why are these two indicators equivalent ?
- What is the formula for defining the "height" output vector when performing agglomerative hierarchical clustering ?

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K-Means Algorithm

The k-means algorithm captures the insight that each point in a cluster should be near to the center of that cluster. It works like this:

- First we fix k , the number of clusters we want to find in the data.
- Then, the centers of those k clusters, called centroids, are initialized in some fashion (randomly).
- The algorithm then proceeds in two alternating parts:
 - In the Reassign Points step, we assign every point in the data to the cluster whose centroid is nearest to it.
 - In the Update Centroids step, we recalculate each centroid's location as the mean (center) of all the points assigned to its cluster.
 - We then iterate these steps until the centroids stop moving, or equivalently until the points stop switching clusters.



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